Master Thesis

Quantifying Enhanced Construct Specification using LLMs

by

Wouter Bakker
(2655935)

First Supervisor: Zhao Yang
Second Reader: Name and Surname

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Wouter Bakker

Vrije Universiteit Amsterdam, Amsterdam w4.bakker@student.vu.nl

Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: LLM · psychometrics · measurement · evaluation

1 Introduction

What determines the quality of a text? And how do we measure it at scale? These two questions have become increasingly important with the rapid development of advanced Natural Language Generation (NLG) systems such as LLMs that enable flexible generation across a wide range of textual tasks [60**?**]. Although Generative AI (GenAI) has significant potential for industrial applications that require automated text generation [42], including marketing copy generation [37], customer-service chatbots [38], and automated translation in for example e-commerce [18], its implementation in business contexts is hindered by the variability of output quality [17]. Without adequate quality control, the accumulation of low-quality output becomes a significant operational problem at high volumes. For content-focused companies, scalable generation therefore shifts the bottleneck from producing text to validating, interpreting, and improving it.

This problem is illustrated by Barter, an influencer marketing platform that facilitates collaborations between companies and content creators. On the Barter platform, companies post collaboration opportunities (*Deals*), to which creators can apply. These deals communicate what is being offered, what is expected from creators in return, and why the collaboration may be attractive. The quality of a deal description is therefore multifaceted: it depends not only on whether the text is fluent or well-written, but also on whether the offer is valuable, the required effort is reasonable, and the pitch is persuasive.

This turns the problem of evaluation into a measurement problem: if the target of evaluation is multifaceted, meaningful measurement requires specifying what construct is being measured, which dimensions constitute it, and how evidence for the resulting score should be interpreted. To formalize this procedure, psychometric frameworks are increasingly being used for NLG evaluation [1, 2, 5, 6, 13, 14, 25, 31, 45, 52, 54]. Psychometrics studies the measurement of latent, unobservable variables through observable indicators. Since the variables of interest in the social sciences are rarely directly observable, meaningful measurement requires specifying how the target construct relates to the indicators or criteria through which it is measured.

Analogously, in text evaluation, complex constructs such as *Text Quality* or *Deal Attractiveness* are not directly readable properties of a text. Rather, they are composite constructs that can be decomposed into theoretically related subdimensions that, enabling more fine-grained diagnostic evaluation. Recent work in NLG evaluation therefore moves beyond holistic scoring by decomposing complex quality constructs into multiple evaluative dimensions. For example, Fabbri et al. [12] decompose *Summary Quality* into *coherence*, *consistency*, *fluency*, and *relevance*, showing that multidimensional evaluation can reveal model-specific failure modes that holistic scores may obscure: pre-trained language models correlate highly with human judgments on consistency and fluency, whereas extractive models struggled with coherence and relevance. Multi-dimensional evaluation has become the dominant paradigm in automated evaluation [62], and with the introduction of LLMs, has become increasingly flexible.

However, many modern evaluation methods do not make their underlying measurement model explicit. As a result, it often remains unclear whether the selected criteria adequately define the target construct, how criterion-level judgments should be recomposed into an overall score, and what claims that score supports. This highlights the central problem of this thesis: for artefact-level evaluation, under what conditions does LLM-based scoring become meaningful measurement?

We argue that artefact-level evaluation is best understood as a formative measurement problem. In a formative measurement model, the target construct is defined by its constituent dimensions. Because of this, omitting or underspecifying a relevant dimension therefore means omitting or changing part of the construct itself [7]. This creates a central validity risk: holistic scores, or decomposed scores that remain weakly specified, may be ill-aligned with the evaluation target.

To test this argument, the thesis compares alternative representations of artefact-level quality. The baseline representation is holistic scoring, where the LLM maps an artefact directly to a single overall score. The formative representation instead measures theoretically specified criteria separately and recomposes them through an explicit regression model. This design addresses two research questions:

RQ1: Does explicit formative recomposition of theoretically specified criteria provide stronger validity evidence than holistic or weakly specified alternatives?

RQ2: Do separately elicited formative criteria support more diagnostically interpretable artefact-level measurements than holistic or weakly specified alternatives?

The analysis is conducted in two complementary validation settings. First, the FeedbackQA dataset [29] is used to assess convergent validity: whether LLM-derived measurements of Q&A answer quality align with human quality judgments. In this setting, *Answer Quality* is decomposed into four theoretically motivated subdimensions that are measured separately in the formative condition. The design also includes an informed holistic control condition, which presents the same four subdimensions to the model but asks for a single overall quality score. This separates the effects of two distinct mechanisms: richer construct specification, where the model is given a more detailed definition of the target construct, and explicit formative recomposition, where the subdimensions are measured independently and recomposed through regression. The comparison therefore tests whether formative measurement improves validity evidence beyond what can be achieved by enhanced construct specification alone.

Second, a real-world dataset from the Barter platform is used to assess predictive validity: whether LLM-derived measurements of deal attractiveness predict observed creator application behavior in an operational influencer-marketing setting. In this setting, *Deal Attractiveness* is represented at three levels: a holistic baseline, a macro-formative decomposition into Reward, Cost, and Pitch Quality, and a fine-grained formative decomposition into ten dimensions. This allows us to test whether explicit formative recomposition improves predictive validity, and whether fine-grained decomposition provides additional validity evidence beyond broad theoretical pillars.

In addition to external validity evidence, the thesis diagnoses the LLM-derived measurements themselves. These diagnostics examine whether the scores exhibit usable scale behavior, whether theoretically distinct criteria remain empirically separable, and whether ratings are sensitive to construct-irrelevant text length. Together, the analyses evaluate whether formative recomposition improves artefact-level LLM measurement not only by strengthening validity evidence, but also by producing criterion-level scores whose interpretation is explicitly tied to the construct being measured.

2 Related Work

This literature review traces two converging developments. The first is the development of automated NLG evaluation from similarity-based metrics to multi-dimensional LLM-based evaluators. The second is the use of language models in the social sciences to measure theoretical constructs from text. These literatures converge on the same methodological problem: LLMs can flexibly produce criterion-level scores, but meaningful measurement requires specifying what construct those scores support, how criteria relate to that construct, and what claims the resulting measurement permits.

2.1 From Similarity Metrics to Multi-dimensional Evaluation

Early methods for automated text evaluation centered on lexical overlap, most prominently the BiLingual Evaluation Understudy (BLEU) [39] and Recall-Oriented Understudy for Gisting Evaluation (ROUGE) [30]. Both metrics operationalize quality through surface-level similarity to reference texts: BLEU measures modified n-gram precision against human translations, while ROUGE measures how much reference content is recovered by a candidate summary.

However, the efficacy of lexical-overlap metrics is limited. Surface-level textual features do not capture the rich semantic information inherent to language, limiting their validity as measures of text quality [43]. This motivated embedding-based evaluation methods such as BERTScore [57], BLEURT [46], and MoverScore [58], which use contextual embeddings to estimate semantic similarity between candidate and reference texts.

Yet, similarity-based metrics remain limited because holistic similarity scores may not capture all relevant information needed to assess the evaluation target. As Gehrmann et al. [19] note, quality differences between advanced NLG models are not fully reflected in surface-level textual features. This motivated more fine-grained evaluation methods that decompose the evaluation target into multiple criteria. A central example is SummEval, where Fabbri et al. [12] decompose "summary quality" into *coherence*, *consistency*, *fluency*, and *relevance*, allowing them to identify distinct model failure modes.

2.2 Automated Multi-dimensional Evaluators

The discussion sparked by Fabbri et al. [12] led to the development of automated multi-dimensional text evaluators. Early studies develop dimension-specific evaluators, such as for *factual consistency* in summarization [27, 50], or *coherence* in dialogue generation [10, 56]. Later studies worked towards unified measurement, such as Zhong et al. [62], who propose a method to quantify any arbitrary dimension using transformer models.

Although their method correlates highly with expert annotations, it requires a costly pre-training stage and reference documents. The introduction of LLMs, known for their general problem-solving capabilities, offered a possible solution to these two problems. Early research into using LLMs as reference-free NLG evaluators [16, 51] exhibited lower correlations with expert judgments compared to neural evaluators such as Zhong et al. [62]. However, with the release of GPT-4, Liu et al. [32] outperformed both reference-based and reference-free baselines by a substantial amount. Zheng et al. [61] introduce novel benchmarks to analyse NLG model performance on open-ended NLG tasks and show that LLM evaluations obtain an agreement with human preferences comparable to human-to-human agreement, solidifying the use of LLMs as general-purpose evaluators.

2.3 Textual Quantification in the Social Sciences

A parallel development has occurred in the social sciences, where researchers also seek to measure abstract constructs from text. This literature is relevant because it treats LLM scoring not merely as prediction or annotation, but as measurement: scores are interpreted in relation to theoretically defined constructs and validated against external evidence.

Given the function of language as a means for expression, linguistic analysis offers a window into human behavior. Text data analysis is used in psychology to study human behavioral traits, such as personality [8], psychological well-being [22], and emotional states [24]; in economics, to study stock market dynamics using text from news and social media [33]; and in political science, to study partisanship using congressional speeches [20]. Given the abundance of text data and the costliness of manual annotation, NLP methods are therefore a natural choice to facilitate the measurement of complex constructs from text.

The social sciences faced limitations similar to those encountered in automated NLG evaluation. Dictionary methods, which consider basic features such as word count, are cheap to implement, but often too superficial to answer complex research questions. Representation-based methods, such as embedding models, are more flexible, but are often best suited to narrower research questions. LLMs again offered the potential for a more general-purpose solution: Rathje et al. [41] show that GPT models can accurately measure numerous psychological constructs from texts in multiple languages, and Le Mens et al. [28] use GPT-4 to quantify semantic properties of texts, with both studies achieving state-of-the-art performance.

2.4 Conditions for Meaningful Measurement

The current LLM-based methods for measuring constructs from text in the social sciences show many similarities to the evaluation of the capabilities of NLG models in ML: Both fields define measurement targets which they operationalize through LLM scoring. However, as a consequence of their measurement assumptions, they differ in the interpretations that the measurements permit.

The considerations in Fabbri et al. [12] are mostly pragmatic. They observed that a single holistic similarity score did not sufficiently cover the quality differences between summarization systems, so they proposed to measure subdimensions that expose these differences. For evaluating models on these specific criteria, this is sufficient. However, their procedure falls short of permitting general claims about summary quality and summarization capabilities.

As Salaudeen et al. [45] argue, this inferential leap requires justifying the relationship between the measurements and the object of the claim; in this case, the relationship between criterion scores and the construct of "summarization ability". This practice is standard in the social sciences: Boyd and Pennebaker [8], Gentzkow et al. [20], Guntuku et al. [22], Kahn et al. [24], Le Mens et al. [28], Manela and Moreira [33], Rathje et al. [41], Yang et al. [55] all operate within a framework where the target constructs are theoretically defined, and where measurements are validated against established measures or theoretically relevant external criteria. Because of this, their measurements are not just criterion-level scores, but construct measurements whose interpretation is anchored in the broader theoretical frameworks of their respective fields.

2.5 Meaningful Multi-Dimensional Measurement

Criticisms of Multi-Dimensional LLM Evaluators To achieve the level of meaningful measurement seen in the social sciences, a growing number of studies are recognizing the need for validity-guided workflows for LLM evaluation based on measurement theory and psychometrics [1, 2, 5, 6, 13, 14, 25, 31, 45, 52, 54].

Yet, advanced benchmarks do not immediately solve the problem of automated evaluation, particularly for open-ended language generation tasks, for which the set of possible correct answers is large and often unknown [15].

As it stands, the focus in automated NLG evaluation still predominantly revolves around theory-agnostic multi-dimensional measurement protocols, where criteria are measured in isolation and the relationship between the measurement and the evaluation target is underspecified. As Arias et al. [3] point out, this holds for 88% of the papers published in NLP evaluation as of 2024, while only 7.6% employ true multicriteria methods, even though linguistic quality is inherently multi-dimensional. This distinction marks the central problem for meaningful multi-dimensional measurement: decomposition is necessary, but not sufficient; additionally, the criteria must be related back to the target construct.

Relating Criteria to Constructs Importantly, this relationship defines the level at which the claims are made: at the system level, or at the artefact level. Benchmarks concern system-level claims, as they attempt to measure some latent model ability. Psychometric methods lend themselves well to this, as many have been developed for this specific purpose. For example, in the same way that latent writing ability can be inferred from grading an essay on multiple criteria, LLMs can be used as scorers to achieve the same goal [53]. However, these methods are not designed for artefact-level claims. Scores that are valid for comparing systems do not automatically support diagnostic claims about individual artefacts. For example, when a model obtains a high score on writing ability, it does not automatically imply that every single text it produces is of high quality, nor does it highlight what needs to be improved about the text. For diagnostic purposes, therefore, different methods are required.

The field has proposed several approaches, with varying measurement assumptions. For automated evaluation of a dialogue generation task, Hashemi et al. [23] propose a method to model rater idiosyncrasies using multi-dimensional LLM scores. They collect data from 24 humans who annotate an average of 31 dialogues on their overall quality. By using criterion-level LLM scores and fitting them to the holistic human ratings using a calibration network, they can accurately model individual rater behavior, enabling them to model for example expert annotators. However, they explicitly take the stance that they model subjective rating behavior, not some ground truth. Kola et al. [26] take a similarly pragmatic approach, where they reuse the same six evaluation criteria for evaluation across multiple distinct evaluation tasks, achieving state-of-the-art alignment with human ratings. They argue that the six-dimensional LLM ratings contain all required information, and a lightweight calibration head is sufficient to extract it.

Meaningful Artefact-Level Diagnostics Hashemi et al. [23] and Kola et al. [26] represent the frontier of multi-dimensional LLM evaluation for diagnostic purposes, and demonstrate that decomposition can improve evaluator performance. However, their utility for artefact-level measurement is limited by the pragmatic orientation of their recomposition mechanisms. Both approaches use calibration

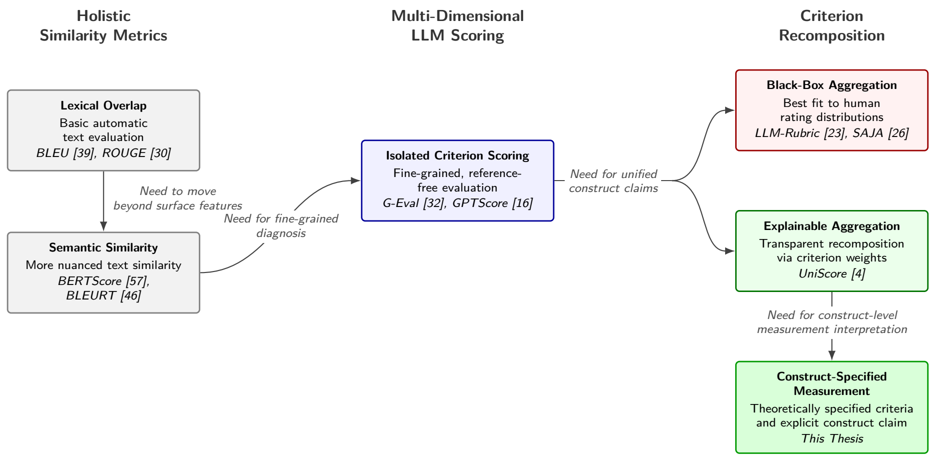

Fig. 1: Lineage of automated NLG evaluation. The field has evolved from holistic similarity metrics toward multi-dimensional LLM scoring. To support artefact-level construct claims, isolated criteria require recomposition. Current recomposition strategies diverge between black-box aggregation, which optimizes alignment with human rating distributions at the expense of explainability, and explainable aggregation, which estimates transparent criterion weights. The present thesis extends the latter path by treating recomposition as a construct-specification problem: theoretically specified criteria are recomposed under an explicit measurement model and interpreted as supporting a construct-level measurement claim.

heads to recombine criterion-level information into a predicted holistic rating. This may improve alignment with human judgments, but it makes the resulting score difficult to interpret diagnostically: even when the predicted holistic score is accurate, it remains unclear which dimensions contributed to the score, and how. Moreover, both approaches explicitly model human idiosyncrasies. This is appropriate when the goal is to reproduce human judgment distributions, since human annotations are often treated as the operational target in alignment research. However, human labels are also inherently noisy [59], and recent evidence suggests that LLMs can sometimes outperform expert coders on difficult annotation tasks [49]. This motivates methods that distinguish alignment with human ratings from meaningful measurement of a target construct, treating human judgments as one source of validity evidence rather than as the object of measurement itself.

Bang et al. [4] take an important step in this direction with UniScore, which derives transparent criterion weights from their discriminative power against an external signal. However, while this makes recomposition interpretable, it leaves the measurement model implicit. Because the criteria remain user-defined, the framework can show how strongly each criterion relates to an external signal, but not whether the selected criteria adequately specify the construct that the final score is interpreted as measuring. This is a central validity risk for artefact-level diagnostics: when an overall score is constructed from multiple criteria, the choice of criteria is itself part of the construct specification.

As summarized in Figure 1, the literature has progressed from holistic similarity metrics to criterion-level LLM scoring and, more recently, to recomposition strategies. What remains unresolved is the measurement-theoretic question of how recomposed criterion scores support artefact-level construct claims.

Taken together, these literatures show that decomposition has become central to both automated NLG evaluation and text-based construct measurement. However, decomposition alone does not establish meaningful measurement. NLG evaluation often emphasizes predictive alignment with human judgments, while social-science measurement emphasizes construct definition and validity evidence. The remaining gap is therefore artefact-level LLM evaluation in which criterion-level scores are not merely produced, but theoretically specified, recomposed, and validated as measurements of a target construct.

The present thesis addresses this problem by treating artefact-level LLM evaluation as a construct-specification problem rather than as just an aggregation problem. It examines whether theoretically grounded criteria, measured separately through LLM prompts and recomposed into an overall score, provide stronger validity evidence and more interpretable artefact-level measurements than holistic or weakly specified alternatives. To formalize this approach, artefact-level scoring is specified as a formative measurement model, drawing on established model specifications from psychometrics.

3 Measurement Framework

This section specifies the validity-guided measurement framework used to evaluate whether LLM-based scores can be interpreted as meaningful measurements. In psychometrics, validity concerns whether theoretical rationales and empirical evidence support the intended interpretation and use of a measurement [35]. Accordingly, the framework has two components:

1. A theoretical account of the measurement model assumed for artefact-level diagnostics;
2. An empirical account of how validity evidence for alternative construct specifications can be evaluated.

Section 3.1 defines the measurement model. Drawing on the distinction between reflective and formative specifications [7], we argue that artefact-level diagnostic evaluation is best understood formatively: the target construct is composed of theoretically relevant dimensions rather than inferred from interchangeable indicators. This makes construct underrepresentation and construct underspecification central validity risks. Section 3.2 then defines the role of empirical validation. Because the target constructs are not directly observable, validity cannot be established by comparison to a known true score. Instead, the analyses assess whether increasingly specified construct representations produce stronger relationships with relevant external criteria: human evaluations in the convergent-validity setting and theoretically related behavioral outcomes in the predictive-validity setting.

3.1 Construct Specification and Measurement Models

LLM-based evaluators become measurement instruments when their scores are interpreted as evidence about a target construct, such as answer quality or deal attractiveness. These constructs are not directly observed in the text, but are specified and operationalized through the measurement procedure used to evaluate it. Such an interpretation necessarily implies a measurement model: the criteria must be assumed to relate to the target construct in a particular way.

Measurement models make this assumption explicit by specifying the direction of the relationship between constructs and their indicators: observed scores may be treated either as effects of an underlying construct or as causal/formative indicators that constitute a composite construct [7, 11]. The measurement model therefore determines how the criteria are assumed to relate to the construct, and what kinds of validity risks follow from that assumption.

Latent Variable Models Latent variable models define the relationship between a target construct and its indicators. Two model specifications are commonly distinguished: formative and reflective [7]. They differ in their assumptions about the direction of the causal relationship between the indicators [1] and the construct.

In formative models, the latent variable is assumed to be a composite construct, caused by the indicators. The canonical example is Socioeconomic Status (SES), which can be defined as the composite of three main features: income, education, and occupational prestige [34]. Formally, for a set of causal indicators indexed by $j = 1, 2, \ldots, J$, we assume

$$\eta = \sum_{j=1}^{J} \gamma_j x_j + \zeta \tag{1}$$

where η is a latent construct, γ_i a formative weight, x_j a causal indicator, and ζ the disturbance term.

In reflective models, the latent variable causes the indicators. For example, following a "g-factor" interpretation of the latent variable intelligence [47], we assume that intelligence is an inherent trait that humans possess which causes people's performance on cognitive tasks. Formally, for any effect indicator j, we assume

$$x_j = \lambda_j \eta + \varepsilon_j \tag{2}$$

[1] We use the term *indicator* here, following standard psychometric vocabulary. In the context of LLM evaluations, this corresponds to the theoretically defined subdimensions, operationalized as LLM-derived criterion scores.

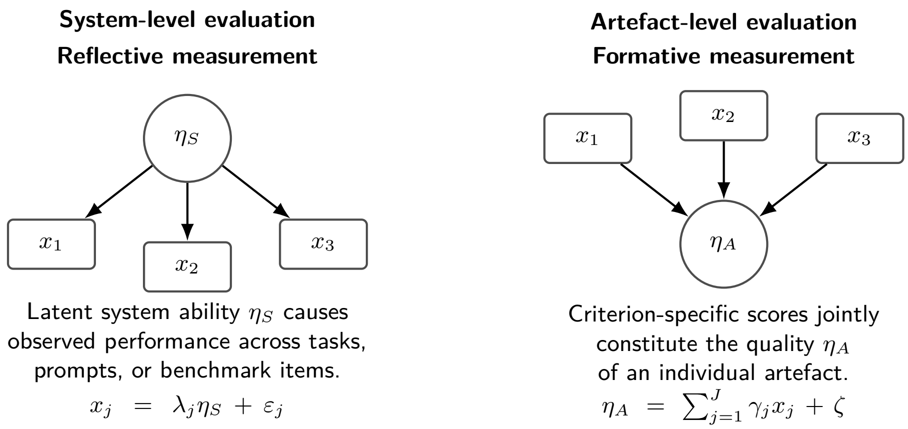

Fig. 2: Basic measurement logic for system-level and artefact-level evaluation as treated in this thesis. In system-level evaluation, observed scores are treated as reflective indicators of an underlying system ability. In artefact-level evaluation, criterion-specific scores are treated as formative components of an artefact-level construct.

where η the latent construct, λ_j a factor loading, x_j is an effect indicator, and ε_j the measurement error.

Model Assumptions & Validity Risks In a formative model, the indicators are not assumed to covary because of a shared latent cause. Returning to the example of SES, highly educated people may have low income, and vice versa, even though education, income, and occupational prestige often covary in practice. This has an important consequence: because each indicator uniquely contributes to the value of the latent variable, omitting an indicator means omitting part of the construct [7]. The main validity risks therefore relate to construct underrepresentation: whether the target construct is adequately covered by the indicators defined.

In contrast, in a reflective model, the indicators are treated as manifestations of a shared latent cause. For example, under a "g-factor" interpretation of intelligence, performance across cognitive tasks is assumed to reflect a common underlying trait. Individual items may contain item-specific error, but with a sufficiently broad set of valid indicators, this error can be averaged out when estimating the latent trait. This illustrates the primary consequence of the reflective assumption: the indicators are theoretically interchangeable. The main validity risks therefore concern whether the indicators can plausibly be treated as manifestations of the same latent construct. If they instead reflect multiple constructs, item-specific effects, or construct-irrelevant sources of variation, the reflective measurement model is misspecified.

Artefact-Level and System-Level Measurement Models As discussed in Section 2.5, system-level evaluation concerns the measurement of model abilities through benchmarks. This resembles a reflective specification: the latent trait is the model ability, while benchmark items, tasks, or criteria function as indicators from which that ability is inferred. Such a model is appropriate when the intended claim concerns average performance across a domain of tasks. However, it does not directly support diagnostic claims about individual artefacts. A model may have high latent writing ability, or score well on a writing benchmark, without every text it produces being high quality.

Artefact-level evaluation therefore requires a different measurement specification. When the target is the quality of a specific answer or the attractiveness of a specific deal, the relevant criteria are not interchangeable indicators of a hidden ability. Rather, they define the dimensions along which the artefact is evaluated. Artefact-level diagnostic evaluation is therefore best understood formatively: the overall construct is composed of theoretically relevant dimensions, and omitting or underspecifying these dimensions risks omitting or misspecifying part of the construct itself.

This distinction has direct implications for artefact-level evaluation. If artefact-level quality is understood formatively, then an overall quality score is only interpretable to the extent that the dimensions constituting that score are specified. A holistic judgement may therefore be useful as a summary assessment, but it leaves the underlying construct representation implicit. In the empirical design, this implication is operationalized by comparing holistic, weakly decomposed, and formative construct representations.

3.2 Construct Validity and Empirical Validation

Because the target constructs are not directly observable, the formative specification cannot be validated by comparing the resulting scores to a known true value. Instead, this thesis adopts a construct-validity approach, where validity concerns whether evidence supports the intended interpretation and use of a measurement [35]. Human ratings and behavioral outcomes are therefore treated as sources of validity evidence, not as the object of measurement itself. They do not exhaust the meaning of answer quality or deal attractiveness, but provide theoretically relevant external criteria against which the expected behavior of the measurements can be assessed.

The role of empirical validation is therefore to evaluate the consequences of construct representation. The previous subsection identified construct underrepresentation and construct underspecification as central validity risks for formative artefact-level measurement. If these risks matter empirically, then more explicitly specified representations of the target construct should show stronger relationships with relevant external criteria than holistic or weakly specified alternatives. Stronger validity evidence is obtained when theoretically specified criterion measurements explain or predict the external criterion more clearly, more strongly, or more consistently than less specified scores.

Let $\mathbf{x}(d)$ denote the vector of LLM-derived criterion measurements for document d, and let $x^{\text{hol}}(d)$ denote a holistic LLM score for the same document. For a validation criterion $y^{(r)}(d)$, the comparison can be written generically as

$$y^{(r)}(d) \approx f_r(\mathbf{x}(d)) \quad \text{vs.} \quad y^{(r)}(d) \approx h_r(x^{\text{hol}}(d)). \tag{3}$$

The empirical question is not whether either side recovers a true score, but whether the more explicit construct representation produces stronger validity evidence for the intended interpretation of the measurement.

This logic is evaluated in two complementary validation settings. Convergent validity is assessed on FeedbackQA, where human quality judgments provide an external reference for Q&A answer quality. Predictive validity is assessed on Barter Deals, where LLM-derived measurements of deal attractiveness are related to theoretically relevant behavioral outcomes, measured as aggregate creator application behavior. The empirical analyses therefore compare the validity evidence produced by holistic, weakly specified, and formative representations of the target construct. The datasets, construct definitions, and model specifications are introduced in Sections 4.3.1 and 4.3.2.

4 Methods

4.1 Research Design and Construct Representations

The previous section motivates artefact-level evaluation as a formative measurement problem. The empirical design tests this claim by comparing alternative construct representations in two complementary validation settings.

FeedbackQA [29] (n = 9,065) is an academic dataset of COVID-19-related Q&A pairs with human evaluations of answer quality. Moreover, for each rating, the humans provide a natural language explanation motivating their rating. Each artefact is a question-answer pair, and the external criterion is the human quality rating assigned to the answer. This provides a convergent-validity setting: LLM-derived measurements of *Answer Quality* are evaluated by testing whether they align with human quality judgments.

Barter Deals (n = 2,960) is an industry dataset of written collaboration opportunities for creators, referred to as *Deals*. In a typical Deal, a company offers a product, service, or experience in exchange for promotional creator content. Creators view these offers on the Barter marketplace and apply to the Deals they find appealing. Each artefact is the written Deal text, consisting of a title, body, and requirements section that together describe the offer and the requested creator deliverables. The external criterion is the number of creator applications received within the first seven days after publication. This provides a predictive-validity setting: LLM-derived measurements of *Deal Attractiveness* are evaluated by testing whether they predict observed creator application behavior.

Across both settings, the central manipulation concerns how the target construct is represented. Holistic baselines ask the LLM to assign a single overall score. Formative representations instead elicit criterion-level scores separately and recompose them through regression. This means that the LLM does not produce the final composite score directly; instead, it produces one score per specified criterion, and the relationship between these criterion-level scores and the external validity criterion is estimated in the corresponding regression model.

Through these experiments, we answer the following research questions:

RQ1: Does explicit formative recomposition of theoretically specified criteria provide stronger validity evidence than holistic or weakly specified alternatives?

RQ2: Do separately elicited formative criteria support more diagnostically interpretable artefact-level measurements than holistic or weakly specified alternatives?

Table 1: Construct representations compared across validation settings. FeedbackQA evaluates *Answer Quality*; Barter Deals evaluates *Deal Attractiveness*. The table summarizes how each target construct is specified in the criterion prompt and how many LLM-derived scores are produced per artefact.

Dataset	**Representation**	**Criterion content**	**Scores / artefact**
FeedbackQA	Naive holistic	Overall answer quality	1
	Informed holistic control	Alignment; Coverage; Concentration; Clarity	1
	Formative	Alignment; Coverage; Concentration; Clarity	4
Barter Deals	Naive holistic	Overall deal attractiveness	1
	Macro-formative	Reward; Cost; Pitch Quality	3
	Fine-grained formative	Ten dimensions nested within Reward, Cost, and Pitch Quality	10

Note. Each score is obtained from one LLM call. Holistic representations produce one overall score, while formative representations produce one score per criterion. Criterion-level scores are recomposed through regression when assessing convergent or predictive validity.

For *FeedbackQA*, *Answer Quality* is decomposed into four dimensions: *Topical Alignment*, *Information Coverage*, *Topical Concentration*, and *Communicative Clarity*. This decomposition is empirically motivated by the natural language explanations provided by crowdworkers and the thematic analysis reported by Li et al. [29], which indicate that answer-quality judgments primarily concern whether answers are relevant, complete, focused, and clearly expressed. The naive holistic baseline evaluates overall answer quality using a minimal construct definition. The informed holistic design control presents the same dimensional content as the formative condition in a single LLM call and asks for one overall score. This separates the effects of two distinct design mechanisms: richer construct specification, where the model is given a more detailed definition of the target construct, and isolated criterion measurement, where the dimensions are elicited separately and recomposed through regression.

For *Barter Deals*, *Deal Attractiveness* is represented through three macro-dimensions derived from the creator's decision problem: *Reward*, *Cost*, and *Pitch Quality*. *Reward* captures what the creator receives from the collaboration, *Cost* captures the effort and constraints required to participate, and *Pitch Quality* captures how clearly and persuasively the opportunity is presented. The naive holistic baseline evaluates overall deal attractiveness using a minimal construct definition. The macro-formative baseline measures the three macro-dimensions separately, reflecting broad multi-dimensional evaluation schemes commonly used in NLG evaluation [12, 32, 62]. The fine-grained formative condition further decomposes the macro-dimensions into ten theoretically specified dimensions: *Brand Prestige*, *Perceived Economic Value*, *Creator Brand Alignment*, *Functional Utility*, *Product Universality*, *Workload Volume*, *Creative Restrictiveness*, *Call-to-Action Strength*, *Rhetorical Objectivity*, and *Proposition Clarity*. This allows the analysis to test whether finer-grained decomposition improves predictive validity beyond broad theoretical pillars. All Barter Deals evaluations are performed on a seven-point Likert scale to allow finer-grained variation in the measurements.

Table 1 summarizes the construct representations compared in each validation setting. Full criterion definitions and scale anchors are provided in Appendix H. Dataset descriptives, text-length distributions, and additional diagnostic plots are reported in Appendix D.1.

4.2 LLM Evaluation Procedure

In the evaluation procedure, the theoretical dimensions defined in Section 4.1 are operationalized as prompt-level criteria. The same procedure is used in both validation settings and is designed to produce criterion-level artefact measurements that can be recomposed in the convergent- and predictive-validity regressions. Because the scoring method relies on access to next-token logits, we use open-weight instruction-tuned models rather than closed API-based systems. This also reflects the operational setting motivating the thesis: repeated artefact-level evaluation requires evaluators that are feasible to run at scale and inspect at the token-probability level. The evaluation models are Llama-3.2-3B-Instruct ("Llama 3.2") [36], Qwen3-4B-Instruct-2507 ("Qwen 3") [48], Qwen/Qwen3.5-4B ("Qwen 3.5") [40], and google/gemma-4-E2B-it ("Gemma 4") [21].

All evaluations use a standardized prompt structure. For artefact d_i, construct representation m, and criterion $c_j \in C_m$, the model input is represented as

$$\mathbf{u}_{ijm} = [\, s \,;\, c_j \,;\, d_i \,],$$

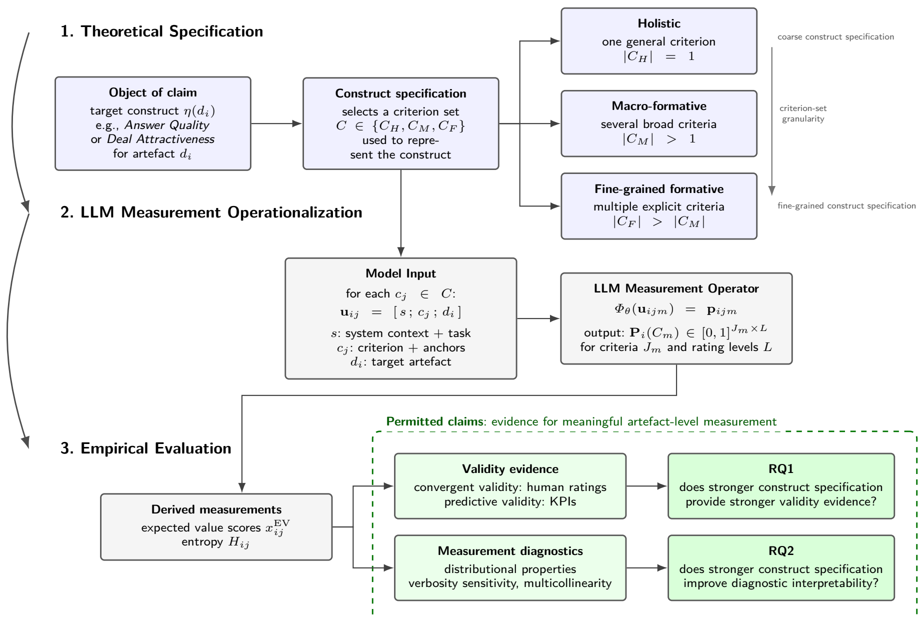

Fig. 3: Validity-guided framework for artefact-level LLM measurement. The framework proceeds from theoretical specification of the object of claim $\eta(d_i)$ to LLM measurement operationalization and empirical evaluation. Construct specification defines a criterion set C, ranging from holistic to macro-formative and fine-grained formative alternatives ordered by criterion-set granularity. For each artefact d_i and criterion $c_j \in C$, the model input $\mathbf{u}_{ij}$ is passed to the LLM measurement operator, which returns a criterion-specific rating distribution $\mathbf{p}_{ij}$. The criterion-level distributions are stacked into $\mathbf{P}_i(C)$, summarized using expected values and entropy, and evaluated through validity evidence and measurement diagnostics. These empirical evidence streams determine the permitted artefact-level claims addressed by RQ1 and RQ2.

where s denotes the system context and task instruction, c_j denotes the criterion specification, and d_i denotes the target artefact. The system context defines the evaluator role and application setting. The criterion specification defines the construct or dimension being measured, including the semantic definition and Likert-scale anchors. The target artefact is the document to be evaluated, such as a Q&A pair in *FeedbackQA* or a Deal text in *Barter Deals*. Across construct representations, s and d_i are held constant; the experimental manipulation is implemented through the content and structure of c_j.

We treat the full scoring procedure as an LLM measurement operator Φ_θ, where θ denotes the parameters of the evaluation model. For each input $\mathbf{u}_{ijm}$, the model first produces next-token logits over the full vocabulary. In implementation, the prompt is formatted using the model-specific chat template, and the assistant turn is prefixed with the literal string `Rating:` to condition the next-token distribution toward the evaluation tokens. A single forward pass then yields the logit vector

$$\mathbf{z}(\mathbf{u}_{ijm}) \in \mathbb{R}^{|\mathcal{V}|},$$

where $\mathcal{V}$ denotes the model vocabulary.

Because the task requires a rating on a bounded Likert scale, we restrict the logits to the valid rating-token set $\mathcal{S} = \{1, \ldots, L\}$, where L is the maximum value of the relevant scale. The restricted logits are converted into a probability distribution over valid rating tokens:

$$P_{\mathcal{S}}(l \mid \mathbf{u}_{ijm}) = \frac{\exp(z_l(\mathbf{u}_{ijm}))}{\sum_{h \in \mathcal{S}} \exp(z_h(\mathbf{u}_{ijm}))}. \tag{4}$$

We denote the resulting rating-token distribution by

$$\mathbf{p}_{ijm} = \{P_{\mathcal{S}}(l \mid \mathbf{u}_{ijm})\}_{l \in \mathcal{S}},$$

so that the composite measurement operator can be written as

$$\Phi_\theta(\mathbf{u}_{ijm}) = \mathbf{p}_{ijm}.$$

For each artefact d_i and construct representation m, the criterion-level distributions can be stacked as

$$\mathbf{P}_i(C_m) \in [0, 1]^{J_m \times L},$$

where $J_m = |C_m|$ is the number of criteria in the representation and L is the number of rating levels.

The primary LLM-derived measurement is the expected value (EV) of the rating-token distribution:

$$x_{ijm}^{\mathrm{EV}} = \sum_{l \in \mathcal{S}} v(l) P_{\mathcal{S}}(l \mid \mathbf{u}_{ijm}), \tag{5}$$

where $v(l)$ maps each rating token to its numeric value. This produces a continuous score that uses the full probability distribution over valid rating categories. EV scores are z-standardized within model, dataset, construct representation, and criterion before entering the convergent- and predictive-validity regressions, so that coefficients are comparable across models and criteria.

In addition to the EV score, we compute the normalized Shannon entropy of the same rating-token distribution as a diagnostic of rating-token dispersion. Entropy indicates whether the EV score is based on probability mass concentrated on one or a few rating tokens or spread across multiple rating tokens. This diagnostic is described in Section 4.4; implementation details are provided in Appendix B.

4.3 Validity Evidence for RQ1

RQ1 evaluates whether explicit formative recomposition of theoretically specified criteria provides stronger validity evidence than holistic or weakly specified alternatives. We assess this in two complementary settings: convergent validity on *FeedbackQA* and predictive validity on *Barter Deals*. In both settings, the alternative construct representations defined in Table 1 are compared by testing how strongly their LLM-derived measurements align with an external validity criterion.

4.3.1 Convergent Validity: FeedbackQA The *FeedbackQA* analysis assesses convergent validity by comparing LLM-derived measurements of *Answer Quality* against human quality ratings. Each Q&A pair has one set of LLM-derived measurements but two human annotations, indexed by $a \in \{1, 2\}$. The regression is therefore estimated at the annotation level: each document-level LLM measurement is paired with each corresponding human rating. Human ratings are treated as continuous interval scores on the original four-point scale.

Let m index the construct representation being evaluated, and let $C_m = \{c_1, \ldots, c_{J_m}\}$ denote the corresponding criterion set. For the naive holistic baseline and informed holistic design control, $J_m = 1$ because each condition produces a single overall score. For the formative condition, $J_m = 4$ because the four answer-quality dimensions are measured separately. For each representation m, we estimate

$$y_{ia}^{\mathrm{H}} = \beta_{0m} + \sum_{j=1}^{J_m} \beta_{jm} x_{ijm}^{\mathrm{EV}} + \delta_m \ln(W_i) + \varepsilon_{iam}, \tag{6}$$

where y_{ia}^{H} denotes the human rating assigned to document d_i by annotator a, x_{ijm}^{EV} denotes the standardized LLM EV score for criterion j under representation m, and W_i denotes the document word count. The log word-count term controls for document-length effects in the relationship between LLM scores and human ratings.

Because the LLM measurements and word counts vary at the document level, while the human ratings vary at the annotation level, residuals may be correlated within documents. We therefore estimate all models using Ordinary Least Squares (OLS) with standard errors clustered at the document level.

Model performance is evaluated using grouped five-fold cross-validation. Folds are grouped by document so that the two annotations for the same Q&A pair never appear in both the training and test sets. For each specification, we generate out-of-fold predictions and compute RMSE, Pearson's r, Spearman's ρ, and Kendall's τ between predicted and observed human ratings. Human-human alignment is reported as a reference benchmark for annotation consistency, not as a competing model.

4.3.2 Predictive Validity: Barter Deals The *Barter Deals* analysis assesses predictive validity by testing whether LLM-derived measurements of *Deal*

Attractiveness predict creator application behavior. Let Y_i denote the number of applications received by Deal i within its first seven days after publication. The outcome is a highly dispersed count variable (mean = 9.48, SD = 16.97, median = 3), with many Deals receiving no applications and a small number receiving very high application counts. We therefore estimate Negative Binomial regression models with a log link. Because multiple Deals may be posted by the same partner, standard errors are clustered at the partner level.

Before model construction, the raw Barter Deals database export is converted into an analysis sample through cleaning and quality filtering. The analysis is restricted to Deals published after April 2025, reflecting exploratory evidence of a structural break after which the platform entered a more stable operational regime for modeling seven-day application behavior. Within this period, Deals are retained only if they have sufficient marketplace exposure, valid substantive text, no low-application duplicate flag, and a valid go-live record. In particular, Deals are required to have at least 160 cumulative exposure hours, ensuring sufficient opportunity to receive applications within the seven-day outcome window. Qualitative inspection indicated that some low-exposure records were test, development, or otherwise non-substantive Deals rather than genuine marketplace offers.

The resulting candidate sample is then transformed into the model matrix used for predictive-validity analysis. The construction of the non-LLM platform controls, the initial quality-filtering procedure, and the model-matrix filtering pipeline are described in Appendix F.

Before adding LLM-derived measurements, we estimate a structured baseline model containing only non-LLM platform variables:

$$\ln(\mu_i) = \beta_0 + \boldsymbol{\gamma}^T \mathbf{Z}_i + \alpha_{\tau(i)}, \tag{7}$$

where $\mu_i = \mathbb{E}[Y_i \mid \mathbf{Z}_i]$ is the expected application count for Deal i. The vector $\mathbf{Z}_i$ contains structural controls for strict follower requirements, competitor density, platform strategy, geographic market scope, and product category. To account for text-length effects, $\mathbf{Z}_i$ also includes quartile indicators for body length and requirements length. Finally, $\alpha_{\tau(i)}$ denotes month-year fixed effects, which absorb platform growth, seasonality, and other time-varying shocks. . The purpose of the baseline model is to partial out predictable market structure before evaluating the additional validity evidence provided by LLM-derived measurements.

For each construct representation m defined in Table 1, the baseline model is augmented with the corresponding LLM-derived scores:

$$\ln(\mu_i) = \beta_{0m} + \sum_{j=1}^{J_m} \beta_{jm} x_{ijm}^{\mathrm{EV}} + \boldsymbol{\gamma}_m^T \mathbf{Z}_i + \alpha_{\tau(i)}. \tag{8}$$

where x_{ijm}^{EV} denotes the standardized LLM EV score for Deal i on criterion j under construct representation m. The number of LLM-derived predictors depends on the representation: $J_m = 1$ for the naive holistic baseline, $J_m = 3$ for the macro-formative representation, and $J_m = 10$ for the fine-grained formative representation.

Predictive-validity evidence is evaluated in two ways. First, we compare in-sample model fit against the structured baseline using log-likelihood, McFadden pseudo-R^2, AIC, and BIC. Improvements are reported relative to the baseline: ΔLL is defined as $LL_{\text{model}} - LL_{\text{baseline}}$, while ΔAIC and ΔBIC are defined as $AIC_{\text{baseline}} - AIC_{\text{model}}$ and $BIC_{\text{baseline}} - BIC_{\text{model}}$, respectively. All three quantities are oriented so that positive values indicate improvement over the structured baseline.

Second, we assess out-of-sample generalization using five-fold cross-validation. In each fold, the model is estimated on the training partition and evaluated on held-out Deals. Folds are grouped by partner so that Deals from the same partner do not appear in both training and test partitions. Predictive performance is summarized using RMSE, Poisson deviance, and Spearman's ρ. RMSE captures prediction error on the original application-count scale, Poisson deviance provides a count-appropriate predictive loss, and Spearman's ρ captures rank-order alignment between observed and predicted application counts.

4.4 Diagnostic Evidence for RQ2

RQ2 concerns whether separately elicited formative criteria support more diagnostically interpretable artefact-level measurements. Since the formative scores are recomposed through regression, the criterion-level measurements should exhibit properties that support coefficient-level interpretation. We therefore assess three diagnostic properties:

Criterion separability. A formative decomposition is diagnostically useful only if theoretically distinct criteria remain empirically distinguishable. This is especially important for regression-based recomposition, where interpreting individual coefficients requires each criterion to contribute unique variation to the outcome. If the criterion scores are highly collinear, the model cannot reliably separate their associations with the external validity criterion. Consequently, the estimated coefficients become statistically unstable, rendering dimension-specific interpretation unreliable. We assess criterion separability using the variance inflation factor (VIF) among criterion-level EV scores. While high VIF values do not necessarily harm aggregate prediction (RQ1), they directly undermine the use of regression weights for diagnostic purposes (RQ2). Details are provided in Appendix E.3.

Distributional behavior. The distributional behavior of the criterion scores is informative for regression-based diagnostics. Since the formative criteria are recomposed through regression, the estimated weights depend on the variation present in the observed LLM-derived scores. Recent work on LLM-based evaluators similarly notes that compressed or imbalanced rubric-score distributions can affect regression-based aggregation and coefficient interpretation [4]. In the present setting, the concern is not only whether scores vary across artefacts, but also how they vary: a model may use the rating scale broadly while still behaving like an ordinal classifier if its probability mass concentrates on a small number of rating tokens.

We therefore inspect the distribution, mean, and standard deviation of the Expected Value (EV) scores together with normalized rating-token entropy. EV dispersion captures *scale utilization*: whether the model spreads artefacts broadly across the rating scale or concentrates scores within a narrow region. Entropy captures *rating-token dispersion*: whether probability mass is concentrated on one or a few rating tokens, producing more quantized EV scores, or spread across multiple rating tokens, producing more continuous EV scores. Together, scale utilization and entropy define *rating regimes*: broad-continuous, broad-quantized, narrow-continuous, and narrow-quantized scoring behavior. These regimes characterize how the evaluator produces criterion scores and therefore how those scores can be used for recomposition and diagnostic interpretation. Detailed interpretations are provided in Appendix E.1.

Residual verbosity sensitivity. Diagnostic criterion scores should reflect the intended construct rather than superficial text properties. Prior work has shown that LLM evaluators may over-reward verbose outputs, making text length a potential source of construct-irrelevant variance [44]. We therefore assess whether EV scores remain sensitive to word count even after restricting the comparison to mature-length texts. For each model and rating target, we compare ratings for documents in the third and fourth word-count quartiles using absolute Cohen's d. Small values indicate that ratings are relatively invariant to additional text length once documents are sufficiently developed, whereas larger values indicate residual verbosity sensitivity. This diagnostic is important because meaningful criterion-level measurement requires the scores to reflect variation in the intended criterion, not construct-irrelevant variation associated with superficial text properties such as length. The detailed implementation is provided in Appendix E.2.

5 Results

The results are organized around the two research questions. The first subsection addresses RQ1 by comparing validation performance across construct representations: holistic baselines, the FeedbackQA informed holistic control, and formative specifications. These comparisons test whether richer construct specification and isolated criterion measurement improve validity evidence. In FeedbackQA, the informed holistic control further distinguishes these two design mechanisms, testing whether separately eliciting the criteria and recomposing them through regression improves validity evidence beyond richer construct specification alone. The second subsection addresses RQ2 by evaluating whether the resulting criterion-level measurements support diagnostic interpretation. It first examines coefficient-level interpretability through criterion-separability diagnostics (VIF), and then evaluates broader measurement behavior through rating regimes and residual verbosity sensitivity.

5.1 Validity Evidence for RQ1

5.1.1 Convergent Validity: FeedbackQA Table 2 reports the main cross-validated convergent-validity results for the EV-only specifications. The full set of cross-validation metrics, including Pearson's r and Kendall's τ, is reported in Appendix G.1. Across all model architectures, the formative condition improves over both the naive holistic and informed holistic baselines on RMSE and Spearman's ρ. The improvements are modest in absolute size, but consistent across models and metrics. The informed holistic condition provides only marginal improvement over the naive holistic condition. This indicates that richer construct specification alone is insufficient: the benefits of the formative specification depend on its combination with isolated criterion measurement and regression-based recomposition.

The strongest convergent-validity results are obtained by the Qwen models. In the formative condition, Qwen 3 and Qwen 3.5 perform nearly identically, reaching Spearman correlations of 0.578 and 0.576, respectively. These values exceed the human-human reference correlation of 0.541, although this benchmark reflects annotation consistency rather than fitted model performance. Llama 3.2 performs weakest across conditions, while Gemma 4 shows intermediate performance.

Table 2: Cross-Validated FeedbackQA Performance by Experimental Condition

Metric	Condition	Gemma 4	Llama 3.2	Qwen 3	Qwen 3.5
RMSE ↓	Holistic Naive	1.119	1.144	1.044	**1.033**
	Holistic Informed	1.111	1.144	1.039	**1.034**
	Formative	1.094	1.128	**1.018**	1.019
Spearman's ρ ↑	Holistic Naive	0.436	0.395	0.544	**0.559**
	Holistic Informed	0.450	0.396	0.550	**0.558**
	Formative	0.480	0.423	**0.578**	0.576
Human-Human Baseline		RMSE = 1.196; Spearman's ρ = 0.541			

Note. Values report out-of-fold cross-validated performance for EV-only specifications. RMSE measures prediction error, while Spearman's ρ measures rank alignment with human quality judgments. Lower RMSE and higher Spearman's ρ indicate better convergent validity. Bold values indicate the best-performing model architecture within each metric and condition. Grey shading indicates the focal formative specification. The human-human baseline compares the two available human annotations and is not a fitted model.

5.1.2 Predictive Validity: Barter Deals Table 3 and Table 4 report the incremental predictive validity of the LLM-derived measurements relative to the structured non-LLM baseline. The central pattern is consistent across in-sample and out-of-sample evaluation: fine-grained formative specifications provide substantially stronger predictive validity than either holistic or macro-formative alternatives. The gains are largest for likelihood-based fit, Poisson deviance, and rank-order performance, while improvements in RMSE are more modest. Across these comparisons, Gemma 4 and Qwen 3.5 obtain the strongest overall results, although the main performance differences are between construct representation rather than between model architectures. Baseline performance is repeated in the table notes as the reference point for these comparisons.

Table 3: In-Sample Model Comparison by Experimental Condition for Barter Deals

Metric	Condition	Gemma 4	Llama 3.2	Qwen 3	Qwen 3.5
	Holistic Naive	0.058	0.056	**0.060**	0.059
Pseudo R^2 ↑	Macro-Formative	0.057	0.058	**0.062**	0.059
	Formative	0.074	0.067	0.072	**0.075**
	Holistic Naive	20.4	2.0	**43.0**	34.5
ΔLL ↑	Macro-Formative	17.4	23.5	**64.8**	33.7
	Formative	207.1	118.7	177.1	**216.2**
	Holistic Naive	38.8	2.0	**84.1**	67.1
ΔAIC ↑	Macro-Formative	28.8	41.0	**123.5**	61.3
	Formative	394.2	217.5	334.1	**412.3**
	Holistic Naive	32.8	-4.0	**78.1**	61.1
ΔBIC ↑	Macro-Formative	10.8	23.0	**105.5**	43.3
	Formative	334.2	157.5	274.2	**352.4**

Note. Values report in-sample Negative Binomial model comparison for EV-only LLM-augmented specifications. The structural baseline has pseudo $R^2 = 0.056$, $LL = -10489.2$, AIC = 21056.4, and BIC = 21290.1. ΔLL, ΔAIC, and ΔBIC are computed relative to this structural baseline and are oriented so that positive values indicate improvement. ΔLL is defined as $LL_{\text{model}} - LL_{\text{baseline}}$, while ΔAIC and ΔBIC are defined as $AIC_{\text{baseline}} - AIC_{\text{model}}$ and $BIC_{\text{baseline}} - BIC_{\text{model}}$, respectively. Bold values indicate the best-performing LLM-augmented model family within each metric-condition row. Grey shading indicates the focal formative specification.

In the in-sample comparison, the formative specifications produce substantially larger improvements in log-likelihood, AIC, and BIC than either the Holistic Naive or Macro-Formative specifications. The contrast is clearest for BIC, where the holistic and macro-formative conditions yield small to moderate improvements, while the formative specifications produce large positive gains across all models. This indicates that fine-grained formative measurements provide incremental predictive information beyond both the structural baseline and the holistic or macro-level construct specifications.

Table 4: Cross-Validated Out-of-Sample Predictive Performance by Experimental Condition for Barter Deals

Metric	Condition	Gemma 4	Llama 3.2	Qwen 3	Qwen 3.5
	Holistic Naive	14.60	14.75	14.46	**14.41**
Poisson Dev. ↓	Macro-Formative	14.64	14.46	**14.38**	14.48
	Formative	**13.23**	13.91	13.53	13.28
	Holistic Naive	19.47	19.52	19.41	**19.35**
RMSE ↓	Macro-Formative	19.51	**19.30**	19.42	19.35
	Formative	18.88	19.08	18.98	**18.86**
	Holistic Naive	0.539	0.530	**0.548**	0.545
Spearman's ρ ↑	Macro-Formative	0.536	0.538	**0.555**	0.539
	Formative	**0.600**	0.574	0.590	0.594

Note. Values report 5-fold cross-validated out-of-sample predictive performance for EV-only LLM-augmented Negative Binomial specifications. The cross-validated structural baseline achieves Poisson Dev. = 14.68, RMSE = 19.48, and Spearman's $\rho = 0.533$. Bold values indicate the best-performing LLM-augmented model family within each metric-condition row. Grey shading indicates the focal formative specification.

The out-of-sample results show the same pattern. Holistic and macro-formative specifications remain close to the structural baseline, whereas the formative specifications improve count-loss and rank-order performance for all models. The strongest cross-validated results are obtained by Gemma 4 and Qwen 3.5 in the formative condition: Gemma 4 achieves the lowest Poisson deviance and highest Spearman correlation, while Qwen 3.5 achieves the lowest RMSE. These results indicate that fine-grained LLM-derived measurements are especially useful for ranking Deals by expected application volume, even though exact count prediction remains difficult.

The weak performance of the macro-formative condition is also informative. Across most comparisons, macro-formative specifications remain close to the holistic baseline and do not approach the performance of the fine-grained formative specifications. This indicates that the predictive gains are not produced by decomposition alone, but by decomposition at a sufficiently fine-grained level. In this setting, broad dimensions such as *Reward*, *Cost*, and *Pitch Quality* appear to underspecify *Deal Attractiveness*, whereas the fine-grained criteria provide the detailed construct specification necessary for substantial gains in predictive validity.

5.2 Diagnostic Evidence for RQ2

5.2.1 Criterion Separability & Coefficient-Level Interpretation

Multicollinearity For *FeedbackQA*, the VIF diagnostics indicate moderate to substantial multicollinearity among the formative criterion scores. All model specifications contain at least three predictors with VIF values above 5, and Llama 3.2 and Qwen 3.5 also contain predictors above 10. This suggests that the four answer-quality criteria are empirically difficult to separate, despite being theoretically specified as distinct formative components. Consequently, the FeedbackQA formative regressions are interpreted at the level of overall model performance and convergent validity, not as evidence for independent criterion-level contributions.

Table 5: Multicollinearity Diagnostics for FeedbackQA Formative Specifications

	VIF diagnostics				
Model	Max	Mean	$N > 5$	$N > 10$	Assessment
Gemma 4	6.75	4.46	3	0	Moderate
Llama 3.2	16.93	9.43	4	2	Substantial
Qwen 3	9.38	6.86	4	0	Moderate
Qwen 3.5	11.39	6.88	4	1	Substantial

Note. Values are computed for the formative criterion-level EV predictors in each model-specific FeedbackQA regression. Max and mean report the maximum and average variance inflation factor across formative predictors. $N > 5$ and $N > 10$ count the number of predictors exceeding the conventional moderate and substantial multicollinearity thresholds, respectively. High values indicate that the formative criteria contain overlapping information, limiting independent coefficient-level interpretation.

For *Barter Deals*, the VIF diagnostics indicate limited multicollinearity across the formative specifications. No fine-grained formative criterion exceeds a VIF of 5 in any model, with maximum VIF values ranging from 1.79 for Gemma 4 to 3.70 for Llama 3.2. The macro-formative specifications show the same pattern, with maximum VIF values ranging from 1.34 for Qwen 3 and Qwen 3.5 to 4.20 for Llama 3.2. The structural controls likewise show no substantial multicollinearity concerns. Full VIF diagnostics for the Barter Deals models, including the baseline controls, are reported in Appendix D.3.

These diagnostics produce a clear contrast between the two validation settings. In FeedbackQA, formative decomposition improves convergent validity, but the criterion scores are too collinear to support dimension-specific coefficient interpretation. In Barter Deals, the fine-grained formative criteria remain sufficiently separable, so the regression weights can be interpreted as diagnostic associations between specific dimensions of *Deal Attractiveness* and creator application behavior.

Coefficient-level interpretation Because the Barter Deals criteria are sufficiently separable, the fine-grained formative coefficients can be interpreted substantively as conditional predictive associations. Table 16 reports the coefficient estimates for the two strongest formative specifications, Gemma 4 and Qwen 3.5. The most robust pattern is the strong positive association of *Product Universality* and *Perceived Economic Value* with creator application behavior. In both models, a one-standard-deviation increase in *Product Universality* is associated with approximately 40% higher expected application counts, while a one-standard-deviation increase in *Perceived Economic Value* is associated with approximately 17–19% higher expected application counts, conditional on the structural controls and the other LLM-derived criteria.

A second consistent pattern is the negative coefficient for *Functional Utility*. This should not be interpreted as evidence that useful products are generally unattractive to creators. Rather, the coefficient is conditional on the other reward dimensions: after accounting for broad consumer appeal, economic value, brand prestige, and creator-brand fit, Deals rated as more utilitarian receive fewer applications. One plausible interpretation is that the residual variation captured by *Functional Utility* reflects more mundane or less content-worthy offers, whereas creators respond more strongly to products that are broadly appealing or economically valuable.

Other dimensions show weaker or less consistent associations. *Brand Prestige* is positive and significant for Gemma 4, but essentially zero for Qwen 3.5, indicating that this association is not robust across evaluator architectures. *Call-to-Action Strength* is positive in both models, but the estimates are small and not statistically significant in the final specifications. The remaining cost and pitch-quality dimensions (see: Appendix H.2) are not consistently associated with application behavior once the reward-related dimensions and structural controls are included. Overall, the coefficient pattern suggests that creator application behavior is primarily driven by the substantive attractiveness of the offered product or service, especially its broad appeal and perceived economic value, rather than by the rhetorical quality of the pitch or the explicit cost dimensions measured here.

5.2.2 Rating Regimes The models exhibit recognizable rating regimes, summarized in Table 6. The table reports two diagnostic quantities: rating-score dispersion, which reflects broad versus narrow use of the rating scale, and rating-token entropy, which reflects continuous versus quantized scoring. Qwen 3.5 shows the most consistently continuous profile, with relatively high entropy in both datasets. Llama 3.2 is also high-entropy, but its Barter Deals ratings are compressed, indicating narrow continuous scoring. Qwen 3 shows the clearest broad-quantized profile, combining broad scale use with low entropy in both

Table 6: Scale Utilization, Rating-Token Entropy, and Implied Rating Regimes

Model	Rating EV: Mean (SD)	Entropy: Mean (SD)	Scale use	Token regime
FeedbackQA (1–4 scale)				
Gemma 4	2.54 (**0.74**)	**0.13** (0.18)	Broad	Quantized
Llama 3.2	2.72 (**0.77**)	**0.72** (0.23)	Broad	Continuous
Qwen 3	2.95 (**0.88**)	**0.12** (0.16)	Broad	Quantized
Qwen 3.5	2.66 (**0.67**)	**0.73** (0.15)	Broad	Continuous
Barter Deals (1–7 scale)				
Gemma 4	3.54 (**1.00**)	**0.30** (0.17)	Broad	Semi-quantized
Llama 3.2	4.29 (**0.40**)	**0.67** (0.08)	Narrow	Continuous
Qwen 3	4.14 (**1.01**)	**0.18** (0.13)	Broad	Quantized
Qwen 3.5	3.68 (**0.79**)	**0.66** (0.09)	Broad	Continuous

Note. Values are averaged across rating targets. Rating EV is the expected-value score computed over the rating-token distribution. Shading highlights the two quantities used to classify regimes: rating SD for broad versus narrow scale use, and entropy mean for continuous versus quantized scoring. Rating means are not directly comparable across datasets because the rating scales differ.

datasets. Gemma 4 is quantized in FeedbackQA but more semi-quantized in Barter Deals.

The contrast between Gemma 4 and Qwen 3.5 in the Barter Deals setting illustrates why this diagnostic matters. The two models obtain comparable predictive-validity results and both use a broad part of the 1–7 scale, indicating substantial between-deal variation for regression-based recomposition. However, Qwen 3.5 expresses this variation through smoother, more continuous criterion distributions, whereas Gemma 4 relies more heavily on quantized, integer-anchored score bands. Thus, similar predictive performance can arise from different measurement regimes. For diagnostic use, Qwen 3.5's smoother criterion-score variation therefore supports more fine-grained criterion-level interpretation, making it better suited to text-revision purposes. Gemma 4 remains useful for diagnostic ranking, but its coefficients are better understood as stepwise differences between lower and higher criterion ratings.

5.2.3 Residual Verbosity Sensitivity Table 7 shows that residual verbosity sensitivity is strongly model-dependent. Llama 3.2 is the only model with systematic medium-sized effects in both datasets: all six FeedbackQA rating targets fall in the medium range, and Barter Deals contains eight medium and one large effect. By contrast, Gemma 4, Qwen 3, and Qwen 3.5 show mostly negligible or small effects, with no medium or large effects in either dataset. This indicates that residual verbosity sensitivity is primarily a model-specific property rather than a rating-target-specific property.

Table 7: Summary of Residual Verbosity-Sensitivity Diagnostics

Dataset	Model	Mean $\|d\|$	Max $\|d\|$	N	Neg./Small/Med./Large
FeedbackQA	Gemma 4	0.16	0.20	6	5/1/0/0
FeedbackQA	Llama 3.2	0.57	0.62	6	0/0/6/0
FeedbackQA	Qwen 3	0.15	0.25	6	4/2/0/0
FeedbackQA	Qwen 3.5	0.11	0.25	6	5/1/0/0
Barter Deals	Gemma 4	0.17	0.46	14	8/6/0/0
Barter Deals	Llama 3.2	0.53	0.85	14	1/4/8/1
Barter Deals	Qwen 3	0.20	0.47	14	8/6/0/0
Barter Deals	Qwen 3.5	0.17	0.43	14	10/4/0/0

Note. Residual verbosity sensitivity is measured as absolute Cohen's d from the Q3–Q4 word-count comparison. Counts report the number of rating targets in the negligible, small, medium, and large effect-size bins, respectively. Smaller values indicate greater invariance to mature-length text differences.

Substantively, these results indicate that Llama 3.2's evaluator scores are more contaminated by residual text-length effects than those of the other models. Because this sensitivity persists after comparing only mature-length texts, it suggests construct-irrelevant variance rather than merely legitimate differences between short and longer artefacts. This weakens the diagnostic interpretability of Llama 3.2's criterion scores: part of their variation appears to reflect verbosity rather than the intended evaluation criteria.

Synthesis Taken together, the diagnostic results indicate that criterion-level LLM measurements cannot be evaluated on validity performance alone. The FeedbackQA formative specifications improve convergent validity, but their high criterion collinearity limits coefficient-level interpretation. In contrast, the Barter Deals formative specifications combine strong predictive validity with sufficiently separable criteria, allowing the regression weights to be interpreted as diagnostic associations. The rating-regime and verbosity diagnostics further show that models with similar validation performance can behave differently as measurement instruments: Gemma 4 and Qwen 3.5 perform similarly in the Barter Deals setting, but differ in rating-token entropy and scale behavior, while Llama 3.2 shows systematic residual verbosity sensitivity despite otherwise usable rating behavior. Overall, RQ2 is therefore answered conditionally: LLM-derived criterion measurements can support meaningful diagnostics, but only when criterion separability, rating behavior, and robustness to construct-irrelevant text features are evaluated alongside convergent or predictive validity.

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A Prompt rendering via `apply_chat_template`

TODO: CHANGE THIS SO IT MATCHES CURRENT PROMPT

We format each prompt using Hugging Face's chat template with `add_generation_prompt=True`, and then append an explicit assistant prefix (shown below). The rendered template is passed to the model, which computes the next-token probabilities after the assistant prefix. The model input is the concatenation of:

(i) system message, (ii) user message, (iii) assistant header inserted by the template, (iv) `assistant_prefix`.

```
<System role header as produced by tokenizer>
Instruction: Rate the Relevance of the Answer provided for
    the Question
on a scale of 1 to 4. Use the following rubric to determine
    the score:

Relevance: Does the answer discuss the specific topic and
    entities requested?

1 = Irrelevant: Discusses a completely different topic (e.g.,
     "baby delivery"
instead of "respirators").

2 = Topic Mismatch: Discusses a related category but the
    wrong specific
entity (e.g., "Work Visa" instead of "Pandemic Visa").

3 = Broadly Relevant: Discusses the correct topic but
    includes significant
tangential or unrelated information.

4 = Precise: Focuses exclusively on the specific entity and
    topic requested
in the question.
</System role header>

<User role header as produced by tokenizer>
Question:
{question}

Answer:
{answer}
</User role header>

<Assistant role header inserted by add_generation_prompt=True
    >
Rating:
```

B Logit-Based Score Extraction

This appendix provides the technical details for the logit-based scoring procedure described in Section 4.2. The procedure converts each prompt-document pair into a continuous Expected Value (EV) score and a normalized entropy diagnostic by restricting the model's next-token distribution to the valid rating tokens.

B.1 Prompt Formatting and Forward Pass

For artefact d_i and criterion c_j, the evaluation input is defined as

$$\mathbf{u}_{ij} = [\, s \,;\, c_j \,;\, d_i \,],$$

where s denotes the system context and task instruction, c_j the criterion specification, and d_i the target artefact. In implementation, this input is formatted using the chat template associated with each model. The system turn contains the evaluator instruction and criterion definition, while the user turn contains the artefact to be evaluated.

The assistant turn is manually prefixed with the literal string `Rating:`. This fixes the answer format and positions the next-token distribution over the possible rating tokens. A single forward pass is then used to obtain the next-token logits

$$\mathbf{z}(\mathbf{u}_{ij}) \in \mathbb{R}^{|\mathcal{V}|},$$

where $\mathcal{V}$ denotes the model vocabulary.

B.2 Restricted Rating-Token Distribution

Let $\mathcal{S} = \{1, \ldots, L\}$ denote the valid rating categories for the relevant Likert scale. For *FeedbackQA*, $L = 4$; for *Barter Deals*, $L = 7$. The logits are restricted to the token IDs corresponding to the valid rating strings. This yields the restricted logit vector

$$\mathbf{z}_{\mathcal{S}}(\mathbf{u}_{ij}) = \{z_l(\mathbf{u}_{ij})\}_{l \in \mathcal{S}}.$$

The restricted logits are converted into a probability distribution by applying a softmax over the valid rating tokens:

$$P_{\mathcal{S}}(l \mid \mathbf{u}_{ij}) = \frac{\exp(z_l(\mathbf{u}_{ij}))}{\sum_{h \in \mathcal{S}} \exp(z_h(\mathbf{u}_{ij}))}. \tag{9}$$

This produces a normalized probability distribution over the rating scale only. Tokens outside the valid rating set are excluded from the scoring distribution.

B.3 Expected Value Scores

The primary LLM-derived measurement is the Expected Value (EV) of the restricted rating-token distribution:

$$x_{ij}^{\mathrm{EV}} = \sum_{l \in \mathcal{S}} v(l) P_{\mathcal{S}}(l \mid \mathbf{u}_{ij}), \tag{10}$$

where $v(l)$ maps each rating token to its numeric value. The EV score therefore uses the full distribution over rating categories rather than only the most likely token. For example, if a model assigns probability mass to both adjacent categories, the resulting EV can fall between integer values.

Before entering the validation models, EV scores are z-standardized within each model, dataset, experimental condition, and criterion:

$$\tilde{x}_{ij}^{\mathrm{EV}} = \frac{x_{ij}^{\mathrm{EV}} - \bar{x}_{\cdot j}^{\mathrm{EV}}}{\mathrm{SD}(x_{\cdot j}^{\mathrm{EV}})}. \tag{11}$$

This standardization ensures that regression coefficients are comparable across criteria and model architectures, despite differences in raw scale use.

B.4 Normalized Entropy

In addition to the EV score, we compute the normalized Shannon entropy of the same restricted rating-token distribution. Let $K = |\mathcal{S}|$ denote the number of valid rating categories. Entropy is defined as

$$H_{ij} = -\frac{1}{\ln(K)} \sum_{l \in \mathcal{S}} P_{\mathcal{S}}(l \mid \mathbf{u}_{ij}) \ln P_{\mathcal{S}}(l \mid \mathbf{u}_{ij}). \tag{12}$$

The normalization by $\ln(K)$ makes entropy comparable across rating scales with different numbers of categories. The statistic takes values in $[0, 1]$: it equals 0 when all probability mass is concentrated on a single rating token and equals 1 when probability mass is uniformly distributed across all valid rating tokens.

Entropy is used as a diagnostic of rating-token dispersion. Low entropy indicates that the EV score is based on concentrated probability mass over one or a few rating tokens, producing more quantized, integer-like scores. Higher entropy indicates that probability mass is distributed more diffusely across the rating scale, allowing EV scores to behave more continuously. The interpretation of entropy together with EV dispersion is discussed in Appendix E.1.

C Sample data

C.1 FeedbackQA examples

Example F01

Question: Is it still possible to access medical appointments and medications through my doctor?
Answer: You can access bulk-billed appointments with your doctor, nurse or mental health professional via phone or videoconference. If you would like to arrange a telehealth appointment, contact your doctor. Accessing medicines while staying at home. Your doctor may be able to arrange to have your prescriptions sent to your pharmacy directly so you can stay at home. For more information on whether you can access this service please call your pharmacy and speak to them about the best way to arrange this.
Rating 1: 4
Feedback 1: Gives information on how patients can access appointments with their doctors during the pandemic.
Rating 2: 3
Feedback 2: This answer relates to accessing medical appointments and medications, but does not give much information.

Table 8: Sampled FeedbackQA items (IDs and labels), and Qwen scores for Relevance, Completeness, Directness, and Holistic criteria

ID	Human label 1	Human label 2	Rel.	Comp.	Dir.	Holistic
F01	4	3	4	3	3	4

C.2 Barter Deals examples

Example B01

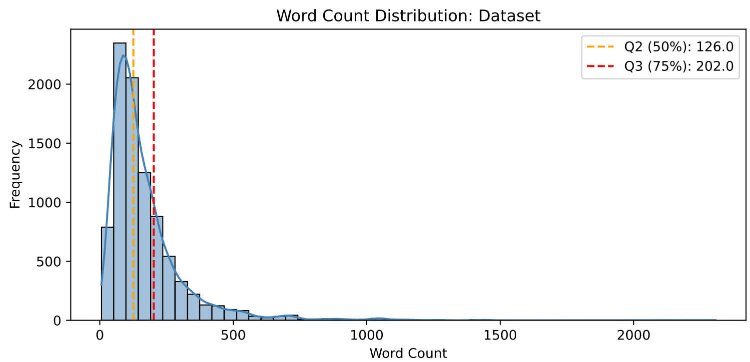
Deal text: We'd love a video capturing the experience of visiting our store in the heart of Amsterdam—from the vibe of the city and the walk over, to exploring the shop itself. While you're there, pick your favorite bag from the collection and make it part of the story!
Keep it authentic, stylish, and feel free to show the charm of the surroundings too!
Applications after 7 days: 39
Product category: Activities

Example B02

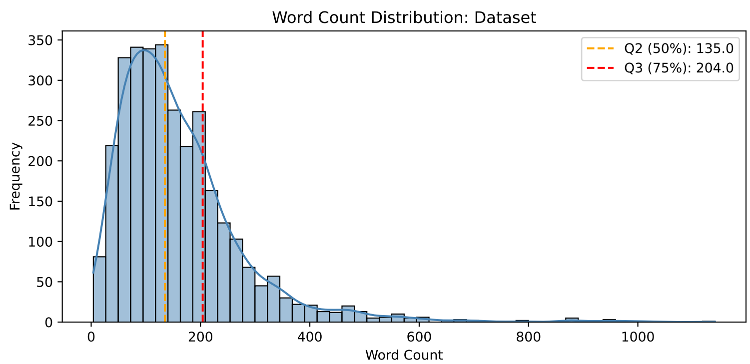
Deal text: Choose 2 coffees and pastries you like:)
Applications after 7 days: 6
Product category: Food

D Additional Descriptives & Diagnostics

D.1 Text-Length Descriptives and Residual Verbosity Sensitivity



(a) FeedbackQA: combined word counts (Question + Answer).



(b) Barter Deals: combined word counts (Title + Body + Requirements).

Fig. 4: Empirical word-count distributions used for the residual verbosity-sensitivity diagnostic. Dashed lines indicate the median and 75th percentile, which define the mature text brackets D_{Q3} and D_{Q4}. For FeedbackQA, $Q(0.50) = 126$ and $Q(0.75) = 202$; for Barter Deals, $Q(0.50) = 135$ and $Q(0.75) = 204$.

D.2 Rating-Target Distributions

This subsection reports the full criterion-level EV rating distributions used to support the aggregate rating-regime diagnostics in the main text.

Table 9: Text-length statistics for Barter Deal text sections

Text Section	Mean Word Count	SD	Median (Q2)	75th Percentile (Q3)
Body	81.89	72.13	64	104
Requirements	74.92	79.55	53	98

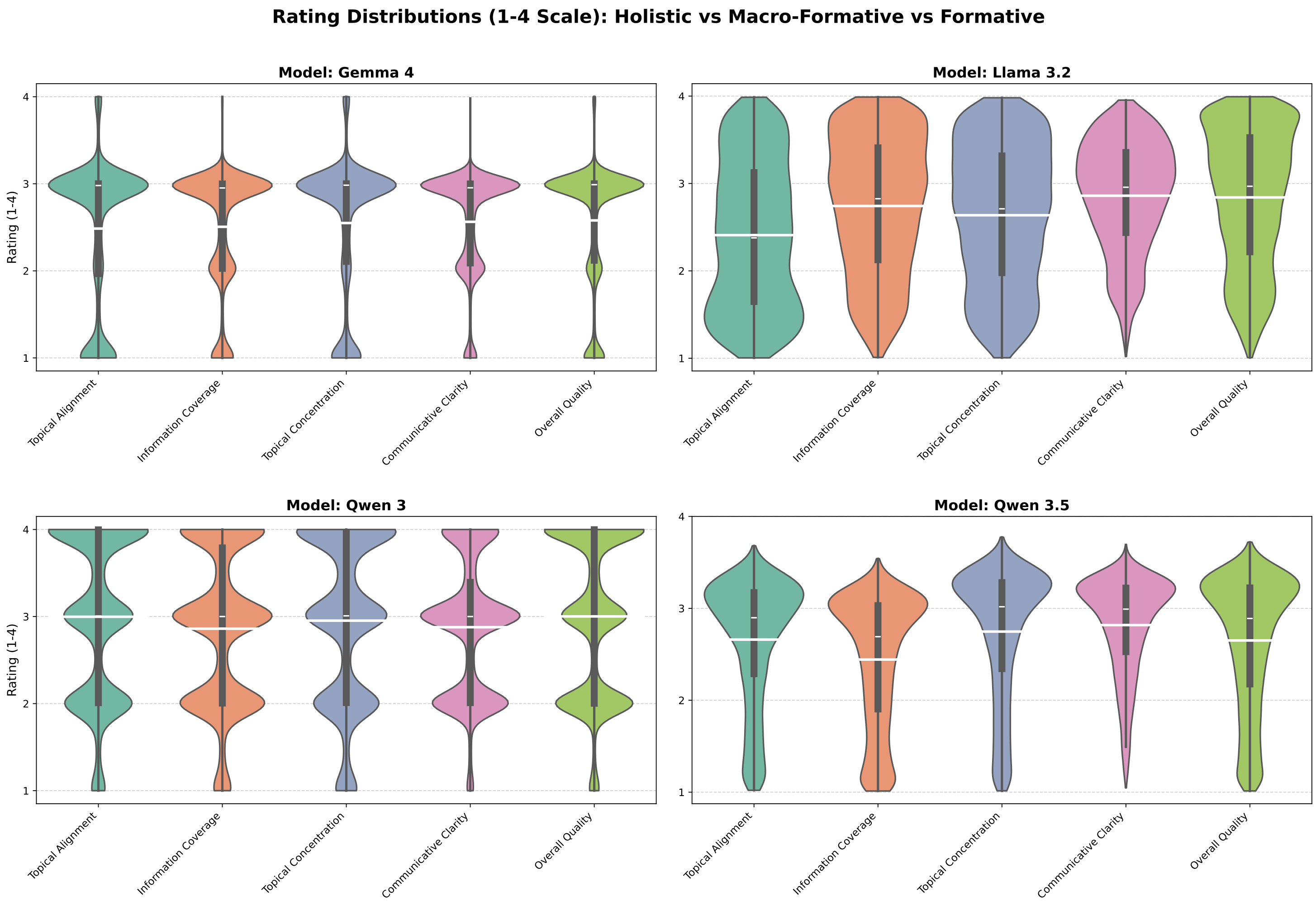


Fig. 5: Rating-target expected-value rating distributions for FeedbackQA.

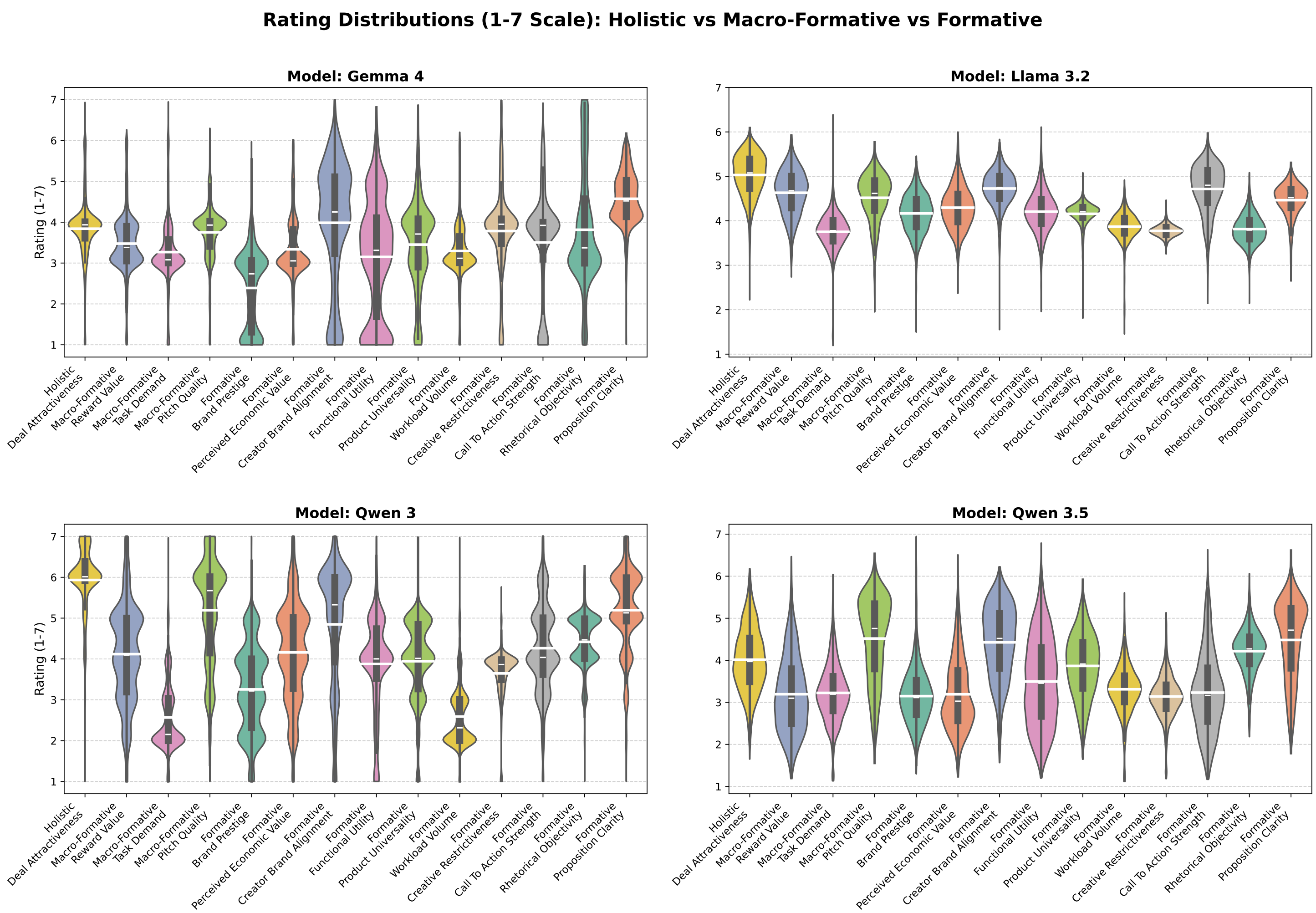


Fig. 6: Rating-target expected-value rating distributions for Barter Deals.

D.3 Additional Multicollinearity Diagnostics

Table 10: VIF Diagnostics for Barter Deals Formative Specifications

Group	Max VIF	Mean VIF	N VIF > 5	N VIF > 10	Multicollinearity
Gemma 4 Formative	1.79	1.36	0	0	Limited
Gemma 4 Macro-Formative	1.70	1.49	0	0	Limited
Llama 3.2 Formative	3.70	2.54	0	0	Limited
Llama 3.2 Macro-Formative	4.20	2.99	0	0	Limited
Qwen 3 Formative	1.97	1.49	0	0	Limited
Qwen 3 Macro-Formative	1.34	1.25	0	0	Limited
Qwen 3.5 Formative	2.07	1.58	0	0	Limited
Qwen 3.5 Macro-Formative	1.34	1.26	0	0	Limited

Note. VIF values are only computed for the LLM-derived evaluations for the Formative and Macro-Formative conditions, and do not include the non-LLM control variables. Values above 5 are interpreted as moderate multicollinearity and values above 10 as substantial multicollinearity.

Table 11: VIF Diagnostics for Barter Deals Baseline Control Specification

Specification	Max VIF	Mean VIF	$N_{\text{VIF}>5}$	$N_{\text{VIF}>10}$	Multicollinearity
Baseline Controls	5.09	2.04	1	0	Moderate

Note. VIF values are computed for the non-LLM control specification used in the Barter Deals predictive-validity models. The only predictor exceeding 5 is *macro_category_Home Cooking & CPG* with a VIF of 5.09. No predictor exceeds 10.

E Supplementary Measurement Diagnostics

This appendix provides additional methodological detail for the diagnostic procedures used to evaluate the LLM-derived measurements. These diagnostics are not themselves validation tests. Instead, they characterize whether the measurements exhibit properties required for their intended use: adequate scale behavior, robustness to residual text-length effects, and empirical separability among formative criteria.

E.1 Interpreting EV and Entropy Regimes

The distributional diagnostics combine two sources of information: the spread of the EV ratings and the entropy of the restricted rating-token distribution. EV spread indicates how much of the available rating scale is used by the model. Entropy indicates whether the EV score is produced by probability mass concentrated on one or a few rating tokens, or by a smoother distribution over multiple rating tokens.

These metrics allow us to distinguish four rating regimes, which serve as interpretive categories used to compare model behavior.

Broad continuous regime. A broad continuous regime occurs when EV ratings show substantial dispersion across the available rating scale and entropy is not uniformly near zero. In this case, the model differentiates between documents while also producing smooth variation between rating tokens. This is the most desirable regime for a logit-derived continuous measurement variable.

Broad quantized regime. A broad quantized regime occurs when EV ratings vary across the scale, but entropy is consistently low. In this case, the model distinguishes between documents, but it does so primarily through near-argmax selection among discrete rating tokens. The resulting scores behave more like hard ordinal classifications rather than continuous measurements.

Narrow continuous regime. A narrow continuous regime occurs when entropy indicates non-degenerate probability distributions, but EV ratings remain concentrated in a restricted part of the scale. In this case, the model produces smooth variation, but within a compressed interval.

Narrow quantized regime. A narrow quantized regime occurs when EV ratings are compressed and entropy is consistently low. In this case, the model uses only a limited part of the rating scale and assigns probability mass to a small number of rating tokens.

E.2 Residual Verbosity Sensitivity

The residual verbosity diagnostic is designed to distinguish general text-length effects from length effects that remain after documents have reached a mature length. Not every association between text length and rating is evidence of verbosity bias. Very short texts may receive lower ratings for construct-relevant reasons, because they may lack the information required to satisfy the criterion. The diagnostic therefore compares documents in the upper half of the length distribution, where additional length is less likely to reflect basic informational adequacy.

Let D represent the corpus of evaluated documents and let $Q(p)$ denote the p-th quantile of the word-count distribution. The third and fourth quartile subsets are defined as:

$$D_{Q3} = \{d \in D \mid Q(0.50) \leq \text{length}(d) \leq Q(0.75)\},$$

$$D_{Q4} = \{d \in D \mid Q(0.75) < \text{length}(d) \leq Q(1.00)\}.$$

A rating target denotes any LLM-derived score produced by a construct representation, including holistic scores, macro-dimension scores, and fine-grained criterion scores. For a given model m and rating traget r_i, let μ_{Q3} and μ_{Q4} denote the mean EV ratings assigned to documents in D_{Q3} and D_{Q4}, respectively. Let s_p denote the pooled standard deviation across the two groups. The signed Q3–Q4 effect is computed as:

$$\delta_{m,r_i} = \frac{\mu_{Q4} - \mu_{Q3}}{s_p}.$$

Positive values indicate that the longest documents receive higher ratings than mature but shorter documents, while negative values indicate that the longest documents receive lower ratings. Because the substantive interpretation of the sign depends on the rating target and its coding [2], the main diagnostic uses the absolute value $|\delta_{m,r_i}|$. This captures the magnitude of residual length sensitivity regardless of direction.

Following Cohen [9], values around 0.2 are interpreted as small, values around 0.5 as medium, and values of 0.8 or above as large.

E.3 Construct Separability and VIF

For the formative specifications, the LLM-derived predictors are intended to represent distinct components of the target construct. Empirical separability is assessed using the Variance Inflation Factor (VIF). For predictor x_j, the VIF is defined as:

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

[2] For example, Llama 3.2 shows a negative Q3–Q4 effect for *Creative Restrictiveness*. This criterion is inversely coded: higher scores indicate more restrictive requirements, whereas for most other criteria higher scores indicate desirable properties. This illustrates why the absolute effect size is used as the main diagnostic.

where R_j^2 is obtained by regressing x_j on the remaining predictors. Higher VIF values indicate that a predictor is strongly predictable from the other predictors, which reduces the interpretability of individual coefficients. VIF values below 5 are treated as indicating limited multicollinearity, values between 5 and 10 as moderate multicollinearity, and values above 10 as substantial multicollinearity. Thus, VIF is used as a diagnostic for coefficient-level interpretability rather than as a test of aggregate predictive performance.

F Barter Deals Control Variables and Model-Matrix Construction

This appendix describes the construction of the non-LLM platform variables used in the Barter Deals predictive-validity models. These variables define the structured baseline model and are included as controls in the LLM-augmented Negative Binomial regressions.

F.1 Engineered Platform Controls

Strict follower requirement. The variable *strict follower requirement* is coded as a binary indicator based on the minimum follower requirement specified for the Deal. Deals with a required minimum follower count above 4,000 are coded as having a strict follower requirement. This control captures eligibility restrictions that may reduce the number of creators who can apply.

Competitor density. Competitor density is computed as the number of other active Deals in the same macro-category on the same publication date. The focal Deal itself is subtracted from the count. This variable controls for same-day category-level competition for creator attention on the marketplace.

Platform strategy. Platform strategy is derived from the Deal title, body text, and creator requirements. These fields are concatenated and searched for platform references using regular expressions for Instagram, TikTok, YouTube, and Facebook. Deals are then grouped into platform-strategy categories: Instagram only, TikTok only, Instagram and TikTok, unspecified, and a residual high-friction deliverable category for more complex or less common platform combinations. In the regression, these categories enter as dummy variables relative to the omitted reference category.

Geographic market scope. Geographic market scope is constructed from the accepted-country information associated with each Deal. The resulting categories distinguish Netherlands-only campaigns, campaigns that include the Netherlands and additional markets, Belgium-only campaigns, Germany campaigns, and other market scopes. These indicators control for differences in the size and relevance of the potential creator pool across geographic markets.

Product category. Product category is constructed from the platform content-type tags and Deal type. Content tags are first cleaned by removing uninformative tags such as UGC, missing values, and null-like entries. Deals are then mapped into macro-categories using rule-based logic. For example, food- and drink-related Deals are separated into *Home Cooking & CPG* or *Dining & Hospitality* depending on whether the Deal is online or physical; entertainment-related Deals are separated into *Digital Entertainment* or *In-Person Experiences* using the same distinction. Other categories include *Beauty*, *Fashion*, *Family & Parenting*, *Lifestyle & Home*, and *Tech & Niche*. Categories with fewer than the minimum required number of observations are reassigned to *Other* for statistical stability.

Text-length controls. Text length is controlled using separate variables for Deal body length and requirements length. These are converted into quartile indicators, allowing the baseline model to account for nonlinear differences between shorter and longer Deal texts without imposing a linear word-count effect. The model also computes the total raw input word count as the sum of title, body, and requirements word counts, which is used for descriptive and diagnostic analyses.

Month-year fixed effects. Publication time is represented using month-year fixed effects derived from the Deal creation timestamp. These fixed effects absorb platform growth, seasonality, and other time-varying shocks shared by Deals published in the same month.

F.2 Sample Construction and Filtering

The sample construction first restricts the raw Barter export to the post-April 2025 analysis period. This restriction reflects exploratory evidence of a structural break around April 2025, after which the platform entered a more stable operational regime for modeling seven-day application behavior. Quality filtering is then applied within this intended analysis period. Deals are required to have at least 160 cumulative exposure hours, ensuring sufficient opportunity to receive applications within the seven-day outcome window. Qualitative inspection indicated that some low-exposure observations were test, development, or otherwise non-substantive records rather than genuine marketplace offers.

Table 12: Barter Deals Sample Construction and Model-Matrix Filtering Pipeline

Filtering step	Start N	End N	Dropped
Initial cleaning and quality filtering			
Raw Deal export	7,967	7,967	–
Restrict to post-April 2025 analysis period	7,967	4,715	3,252
Require at least 160 cumulative exposure hours	4,715	3,544	1,171
Remove short-text/spam candidates	3,544	3,543	1
Remove Deals flagged as duplicate	3,543	3,538	5
Remove Deals that never went live	3,538	3,538	0
Model-matrix preparation			
Cleaned candidate Deals	3,538	3,538	–
Remove physical Deals without company location	3,538	3,411	127
Collapse low-N categories	3,411	3,411	0
Remove incomplete recent weeks	3,411	3,302	109
Remove anomalous high-volume zero-seven-day Deals	3,302	3,250	52
Remove archived Deals	3,250	3,226	24
Apply maximum follower-requirement threshold	3,226	3,066	160
Apply minimum seven-day applications filter	3,066	3,066	0
Apply minimum total applications filter	3,066	3,066	0
Remove invalid Deals	3,066	3,066	0
Remove excluded legacy partners	3,066	2,996	70
Remove excluded partners	2,996	2,962	34
Apply configured log transformations	2,962	2,962	0
Final model-ready sample	7,967	2,962	5,005

Note. The table reports the sequential filtering steps used to construct the Barter Deals model matrix. The initial cleaning stage standardizes the raw export, restricts the analysis to the post-April 2025 operational period, removes low-exposure Deals, and excludes obvious non-substantive records. The model-matrix preparation stage constructs engineered controls and applies additional exclusions before estimation. The final model-ready sample contains 2,962 Deals, retaining 37.2% of the full raw export, 62.8% of the post-April candidate set, and 83.7% of the cleaned candidate sample. Category reassignment and log-transformation steps do not remove observations.

G Additional Regression Results

G.1 Full FeedbackQA Cross-Validation Results

EV-only model results Table 13 reports the full set of cross-validated convergent-validity metrics for the FeedbackQA analysis. The main text reports the compact comparison using RMSE and Spearman's ρ only; this appendix table additionally reports Pearson's r and Kendall's τ. The full results confirm the same substantive pattern: the formative condition consistently outperforms the holistic baselines across model architectures, while the informed holistic condition provides only marginal gains over the naive holistic condition.

Table 13: Full Cross-Validated Convergent Validity Performance on FeedbackQA

Model	Condition	RMSE ↓	Pearson's r ↑	Spearman's ρ ↑	Kendall's τ ↑
Gemma 4	Holistic Naive	1.119	0.440	0.436	0.336
Gemma 4	Holistic Informed	1.111	0.454	0.450	0.347
Gemma 4	Formative	1.094	0.479	0.480	0.370
Llama 3.2	Holistic Naive	1.144	0.396	0.395	0.302
Llama 3.2	Holistic Informed	1.144	0.397	0.396	0.302
Llama 3.2	Formative	1.128	0.424	0.423	0.323
Qwen 3	Holistic Naive	1.044	0.546	0.544	0.421
Qwen 3	Holistic Informed	1.039	0.553	0.550	0.426
Qwen 3	Formative	1.018	0.577	0.578	0.449
Qwen 3.5	Holistic Naive	1.033	0.559	0.559	0.435
Qwen 3.5	Holistic Informed	1.034	0.558	0.558	0.434
Qwen 3.5	Formative	1.019	0.576	0.576	0.449
Human-Human	Baseline	1.196	0.542	0.541	0.467

Note. Values report out-of-fold cross-validated performance for EV-only specifications. RMSE measures prediction error; Pearson's r, Spearman's ρ, and Kendall's τ measure alignment with human quality judgments. Lower RMSE and higher correlation coefficients indicate better convergent validity. The human-human baseline compares the two available human annotations and is included as a reference point for annotation consistency, not as a fitted model.

G.2 Barter Deals Regression Results

Structural baseline model results (excluding LLM evaluations)

Structural baseline coefficient estimates

Selected fine-grained formative coefficient estimates [Gemma 4 and Qwen 3.5 coefficient table]

Table 14: Structural Baseline Model Performance for Barter Deals

Evaluation	Metric	Value
In-sample fit	Pseudo R^2	0.056
	LogLik	-10489.2
	AIC	21056.4
	BIC	21290.1
Cross-validated performance	RMSE	19.48
	Poisson deviance	14.68
	Spearman's ρ	0.533

Note. The structural baseline is a Negative Binomial model containing only non-LLM platform covariates, including the controls and fixed effects specified in the methods section. In-sample fit statistics provide the reference point for the likelihood-based LLM comparisons in Table 3; cross-validated performance metrics provide the reference point for the out-of-sample comparisons in Table 4.

Model specification notes [Short note: controls, fixed effects, clustered SEs, standardized predictors]

Table 15: Selected Structural Baseline Coefficients for Barter Deals

Predictor group	Predictor	β	SE	IRR
Deal controls	Competitor density	-0.042^{***}	0.008	0.959
Deal controls	Strict follower requirement	-0.172^{**}	0.060	0.842
Market scope	Non-Dutch international	-1.325^{***}	0.245	0.266
Market scope	Netherlands only	-0.392^{***}	0.066	0.676
Market scope	Belgium only	-1.295^{***}	0.185	0.274
Market scope	Germany	-1.810^{***}	0.084	0.164
Market scope	Other	-1.434^{***}	0.156	0.238
Product category	Beauty	0.602^{**}	0.191	1.826
Product category	Digital Entertainment	−0.052	0.201	0.949
Product category	Dining & Hospitality	0.271	0.195	1.311
Product category	Family & Parenting	−0.116	0.223	0.890
Product category	Fashion	0.512^{*}	0.202	1.669
Product category	Home Cooking & CPG	$0.353^{\dagger}$	0.187	1.423
Product category	In-Person Experiences	−0.302	0.242	0.740
Product category	Lifestyle & Home	0.579^{**}	0.197	1.784
Product category	Tech & Niche	−0.253	0.256	0.777
Product category	Uncategorized	−0.059	0.261	0.943
Platform strategy	Instagram and TikTok	0.302^{*}	0.127	1.353
Platform strategy	Instagram only	0.282^{*}	0.140	1.326
Platform strategy	TikTok only	0.343^{*}	0.136	1.409
Platform strategy	Unspecified	0.305^{*}	0.124	1.357
Requirements length	Medium-short	−0.057	0.077	0.944
Requirements length	Medium-long	−0.058	0.089	0.944
Requirements length	Long	−0.084	0.090	0.920
Body length	Medium-short	−0.045	0.067	0.956
Body length	Medium-long	−0.073	0.072	0.930
Body length	Long	−0.107	0.077	0.898

Note. Coefficients are from the structural baseline Negative Binomial regression with log link. The table reports substantive non-LLM covariates; month-year fixed effects, the intercept, and the dispersion parameter are included in the fitted model but omitted from the table. SE denotes the clustered standard error. IRR denotes the incidence-rate ratio, $\exp(\beta)$; values above 1 indicate higher expected application counts and values below 1 indicate lower expected application counts, conditional on the other covariates. Categorical predictors are interpreted relative to the omitted reference category in the fitted design matrix. Standard errors are clustered at the partner level. Significance levels: $^{\dagger}p < .10$, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Table 16: Fine-Grained Formative Coefficient Estimates for the Two Strongest Barter Deals Specifications

	Gemma 4 Formative		Qwen 3.5 Formative	
Predictor	β	IRR	β	IRR
Brand Prestige	0.077^{**}	1.080	0.000	1.000
	(0.027)		(0.029)	
Call-to-Action Strength	0.040	1.041	0.051	1.052
	(0.032)		(0.034)	
Creative Restrictiveness	-0.019	0.981	-0.039	0.962
	(0.027)		(0.031)	
Creator Brand Alignment	0.038	1.038	0.030	1.030
	(0.032)		(0.034)	
Functional Utility	-0.142^{***}	0.867	-0.064^{*}	0.938
	(0.029)		(0.031)	
Perceived Economic Value	0.171^{***}	1.187	0.155^{***}	1.167
	(0.028)		(0.029)	
Product Universality	0.338^{***}	1.403	0.337^{***}	1.400
	(0.032)		(0.029)	
Proposition Clarity	-0.020	0.980	-0.006	0.994
	(0.026)		(0.037)	
Rhetorical Objectivity	-0.028	0.973	-0.048	0.954
	(0.030)		(0.029)	
Workload Volume	-0.011	0.989	0.018	1.018
	(0.030)		(0.035)	

Note. Coefficients are from Negative Binomial regressions with log link. The table reports the two strongest fine-grained formative specifications: Gemma 4, which achieves the best cross-validated Poisson deviance and Spearman correlation, and Qwen 3.5, which achieves the strongest in-sample fit and lowest cross-validated RMSE. LLM-derived predictors are z-standardized EV scores. Each coefficient therefore represents the expected change in log application count associated with a one-standard-deviation increase in the corresponding criterion score, conditional on the structural controls and month-year fixed effects. IRR denotes the incidence-rate ratio, $\exp(\beta)$, which expresses the same coefficient on the expected-count scale. For example, an IRR of 1.40 indicates approximately 40% higher expected applications, while an IRR of 0.87 indicates approximately 13% lower expected applications, holding the other variables constant. Structural controls and fixed effects are included in the estimation but omitted from the table. Standard errors, reported in parentheses, are clustered at the partner level. Significance levels: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

H Evaluation Criteria and Rating Anchors

H.1 FeedbackQA Criteria

Table 17: Formative dimensions of *Q&A Answer Quality*

Dimension	Definition
Topical Alignment	The degree to which the answer addresses the issue and stated constraints of the question.
Information Coverage	The degree to which the answer provides enough relevant information to serve as a full answer.
Topical Concentration	The degree to which the answer remains centered on the asked issue throughout the passage.
Communicative Clarity	The degree to which the answer is expressed in a clear, well-organized, and easy-to-follow way.

H.2 Barter Deals Criteria

Table 18: Fine-grained dimensions of *Deal Attractiveness* mapped to the macro-formative pillars

Dimension	Pillar	Definition
Brand Prestige	Reward	How well-known and trusted the brand name is in the market.
Perceived Economic Value	Reward	The estimated retail value and premium nature of the product, service, or compensation offered to the creator.
Creator Brand Alignment	Reward	How naturally the product, service, or experience fits within a professional content creator's social media presence.
Functional Utility	Reward	The practical, everyday usefulness and utilitarian value of the physical product or service being offered.
Product Universality	Reward	The mass-market consumer appeal of the physical product or service on offer.
Workload Volume	Cost	The total time and effort required to execute the collaboration.
Creative Restrictiveness	Cost	The level of brand control over the creator's creative process.
Call-to-Action Strength	Pitch Quality	The level of excitement and warmth in the brand's invitation to collaborate.
Rhetorical Objectivity	Pitch Quality	The level of factual realism versus promotional marketing hype in the pitch.
Proposition Clarity	Pitch Quality	The structural clarity and cognitive ease of processing the pitch.